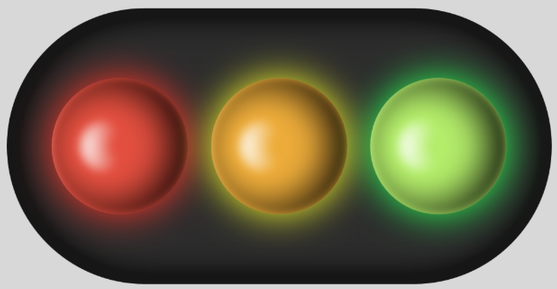
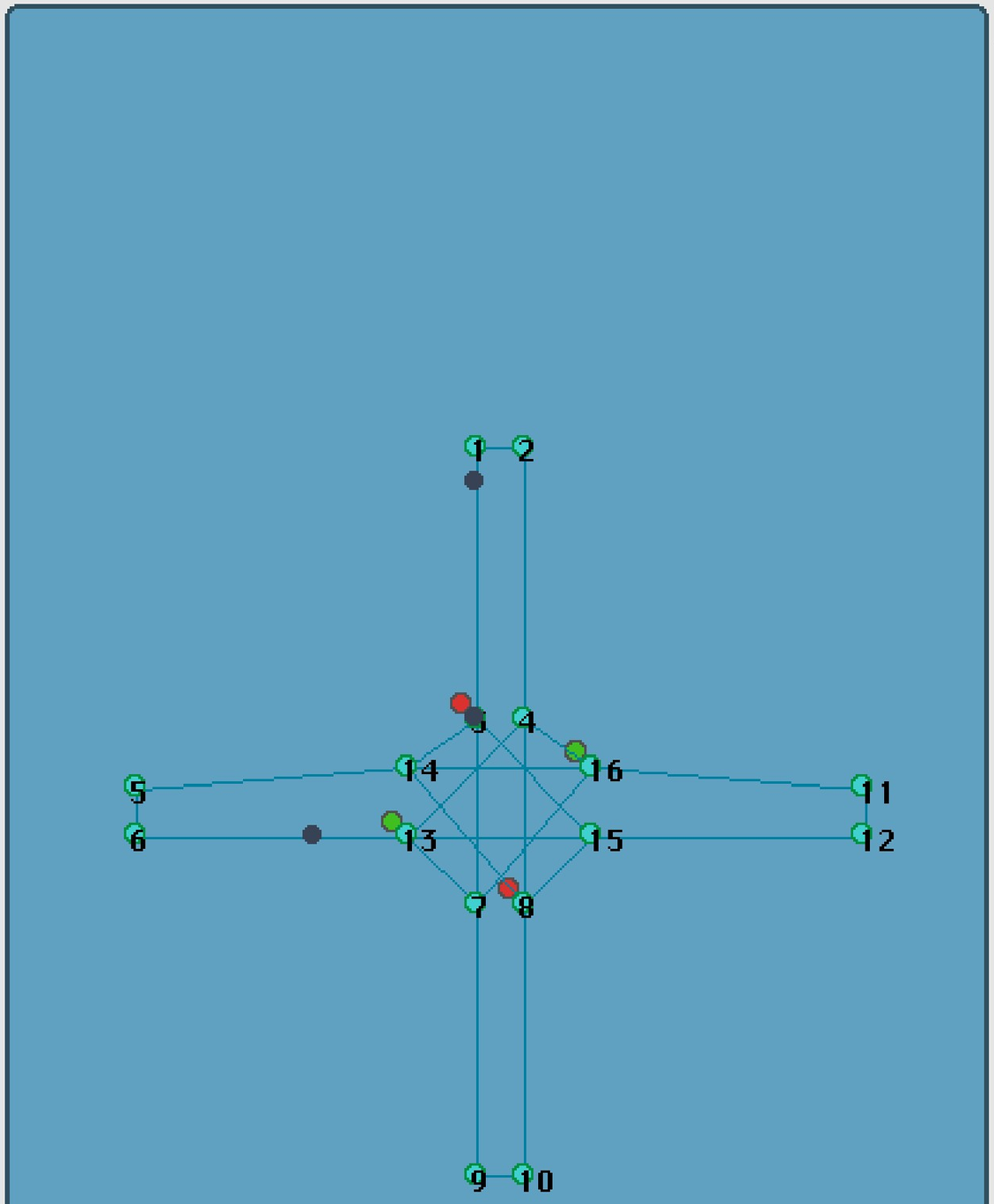
Motivation

The web has made computing more accessible to more people, but web platforms hand control to large corporations.
The latest tech should be online yet running on-device, not in datacenters. Reinforcement learning helps find solutions to hard problems. City traffic causes accidents, pollution, and lost time. Light cycles are one part of the issue a custom learning model can help solve.

Innovation

We wrote CityPulse from scratch in C, with no dependencies, to put us in control. Our custom proximal policy optimization (PPO) model works with a dynamic vehicle simulation based on real data. These operate on a customizable graph structure that can match real road networks.





Implementation

CityPulse is written from scratch in C with no libraries, not even libc. We wrote a build system with code parsing and metaprogramming, a UI system, and many other modules.
Freedom from libraries puts us in control resulting in a small and simple, yet fast and lean app, that works exactly the way we want it to.

